

Determination of citrulline in watermelon rind.



Watermelon (*Citrullus vulgaris* Schrad.) is a natural and rich source of the non-essential amino acid citrulline. Citrulline is used in the nitric oxide system in humans and has potential antioxidant and vasodilatation roles. A method using gas chromatography-mass spectrometry (GC-MS) was developed to separate citrulline from glutamic acid, which co-elute when analyzed by high performance liquid chromatography. Watermelons were analyzed by GC-MS to determine the citrulline content among varieties, types, flesh colors, and tissues. Citrulline content ranged from 3.9 to 28.5 mg/g dry weight (dwt) and was similar between seeded and seedless types (16.6 and 20.3 mg/g dwt, respectively). Red flesh watermelons had slightly less citrulline than the yellow or orange flesh watermelons (7.4, 28.5 and 14.2 mg/g dwt, respectively). Rind contained more citrulline than flesh on a dry weight basis (24.7 and 16.7 mg/g dwt, respectively) but a little less on a fresh weight (fwt) basis (1.3 and 1.9 mg/g fwt, respectively). These results indicate that watermelon rind, an underutilized agricultural waste, offers a source of natural citrulline.